

NGUYEN KHAC BAO

AI Engineer Intern / Computer Vision / Dataset & Quality Assurance

Ho Chi Minh City • onekingdom564@gmail.com • GitHub

Education

Posts and Telecommunications Institute of Technology (PTIT)

Ho Chi Minh City, Vietnam

B.S. in Information Technology

2022 – Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision

Technical Skills

Languages: Python, C/C++

MLOps / Deploy: Docker, Docker Compose, Vast.ai GPU

Frameworks: PyTorch, Ultralytics YOLO, OpenCV, FastAPI

Tools: Linux, Git, Jupyter, Google Colab, Roboflow

AI/CV Expertise: Object Detection, Tracking, Segmentation, OCR, Keypoint Detection

English: Reading technical documentation proficiently

Dataset QA: metadata CSV/manifest, annotation review, train/val/test split, invalid bbox checks, duplicate review, blur/dark/occlusion flags, changelogs

Projects

FallGuard Labeling Tool - Computer Vision Dataset Assistant

2026

Main portfolio project for dataset labeling and QA workflows **Tech:** Python, Streamlit, OpenCV, YOLO-Pose, Pandas |

Source code: GitHub Repository

- Built an end-to-end dataset assistant: raw image/video ingestion → metadata tracking → auto pose labels → human review → QA reports → reviewed-only export.
- Implemented a review UI with bounding boxes, 13 keypoints, confidence, label correction, uncertain/exclude actions, and manual keypoint editing.
- Processed 5 videos into 656 frames and exported annotations, keypoints, event logs, QA reports, version snapshots, and validation checks.

Real-Time Face Mask Compliance Detection System

2026

End-to-end real-time CV system with dataset conversion, training, and demo **Tech:** YOLOv8, FastAPI, WebSocket, OpenCV, Docker, Vast.ai | **Source code:** GitHub Repository

- Trained a 3-class detector for `correct_mask`, `incorrect_mask`, and `no_mask`; built live webcam inference with FPS, latency, counts, and violation snapshots.
- Converted PWMFD from VOC XML to YOLO format: 9,205 images and 18,528 labeled face instances with class mapping, split validation, empty-label handling, and audit reports.
- Trained YOLOv8n on Vast.ai RTX 3090; achieved validation mAP@50 0.970 and mAP@50:95 0.749, while documenting the limited `incorrect_mask` sample size.
- Added real-world test capture metadata for mask subtype, lighting, occlusion, blur, and reflection to support edge-case analysis.

Traffic Violation Detection & License Plate Recognition

2025

Fixed-camera video analytics and dataset QA project **Tech:** YOLO, OpenCV, OCR, ByteTrack, Python, JSON evidence pipeline | **Source code:** GitHub Repository

- Built a traffic video pipeline for vehicle detection/tracking, traffic-light ROI state estimation, stop-line crossing logic, violation events, and plate OCR.
- Prepared a compact CCPD subset with 8,598 images, YOLO plate labels, OCR crops, train/val/test splits, negative samples, and manifest metadata.
- Added audit artifacts for missing files, invalid YOLO boxes, duplicate candidates, dark/heavy-blur/small-plate flags, and contact sheets for human QA.